



SIMATS ENGINEERING

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Course Code: MLA0408

Slot: C

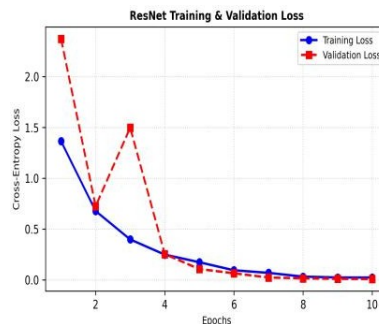
Course Name: DEEP LEARNING

Course Faculty: Dr D Sudhagar

Project Title: Deep Learning-Based Smart Real-Time Emergency Resource Allocation and Inventory Monitoring System

MODULE : Emergency Request & Resource Demand

Module Workflow: Traffic Sign Feature Extraction & Classification Engine



Test Confusion Matrix (ResNet)

	C0	C1	C2	C3	C4	C5	C6	C7	C8	C9
C0	40	0	0	0	0	0	0	0	0	0
C1	0	40	0	0	0	0	0	0	0	0
C2	0	0	40	0	0	0	0	0	0	0
C3	0	0	0	40	0	0	0	0	0	0
C4	0	0	0	0	40	0	0	0	0	0
C5	0	0	0	0	0	40	0	0	0	0
C6	0	0	0	0	0	0	40	0	0	0
C7	0	0	0	0	0	0	0	40	0	0
C8	0	0	0	0	0	0	0	0	40	0
C9	0	0	0	0	0	0	0	0	0	40

Project Description:

- Collected real-time emergency requests and identified the resources required for each situation.
- Classified emergency requests based on urgency, type, location, and resource requirements.
- Analyzed request patterns to understand resource demand during different emergencies.
- Used deep learning techniques to predict the type and quantity of resources needed.
- Developed a prioritization approach to identify critical requests and support faster response

Key Components

My contribution focuses on the Emergency Request & Resource Demand module of the system. I worked on organizing real-time emergency requests, identifying the required resources, analyzing demand patterns, and supporting demand prediction using deep learning techniques. The module is designed to prioritize critical emergencies and provide reliable resource-demand information to the allocation module, helping the overall system achieve faster and more efficient emergency response.

Student Signature

Guide Signature